

ORIGINAL ARTICLE

Irregular sleep and event schedules are associated with poorer self-reported well-being in US college students

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Data collection and processing were performed at MIT and the Brigham and Women's Hospital/Harvard Medical School (BWH/HMS), Boston, USA. Data analysis was performed at the BWH/HMS.

Abstract

Study Objectives: Sleep regularity, in addition to duration and timing, is predictive of daily variations in well-being. One possible contributor to changes in these sleep dimensions are early morning scheduled events. We applied a composite metric—the Composite Phase Deviation (CPD)—to assess mistiming and irregularity of both sleep and event schedules to examine their relationship with self-reported well-being in US college students.

Methods: Daily well-being, actigraphy, and timing of sleep and first scheduled events (academic/exercise/other) were collected for approximately 30 days from 223 US college students (37% females) between 2013 and 2016. Participants rated well-being daily upon awakening on five scales: *Sleepy–Alert*, *Sad–Happy*, *Sluggish–Energetic*, *Sick–Healthy*, and *Stressed–Calm*. A longitudinal growth model with time-varying covariates was used to assess relationships between sleep variables (i.e. CPD^{sleep}, sleep duration, and midsleep time) and daily and average well-being. Cluster analysis was used to examine relationships between CPD for sleep vs. event schedules.

Results: CPD for sleep was a significant predictor of average well-being (e.g. *Stressed–Calm*: $b = -6.3$, $p < 0.01$), whereas sleep duration was a significant predictor of daily well-being (*Stressed–Calm*, $b = 1.0$, $p < 0.001$). Although cluster analysis revealed no systematic relationship between CPD for sleep vs. event schedules (i.e. more mistimed/irregular events were not associated with more mistimed/irregular sleep), they interacted upon well-being: the poorest well-being was reported by students for whom both sleep and event schedules were mistimed and irregular.

Conclusions: Sleep regularity and duration may be risk factors for lower well-being in college students. Stabilizing sleep and/or event schedules may help improve well-being.

Statement of Significance

With mental health problems rising among college students, negative well-being is a cause of concern. Emerging evidence suggests a link between irregular sleep and worse well-being. A potential driver of irregular sleep is irregularly scheduled morning-events (e.g. classes), but this relationship has not been carefully investigated. We studied sleep, scheduled events, and well-being in undergraduates using a recently developed metric for quantifying irregular patterns on different timescales (daily, monthly). We discovered that irregular sleep predicts lower average well-being and short sleep predicts lower daily well-being. Surprisingly, we found no association of irregular sleep with irregular events but both factors combined associate with worse well-being. Future work should investigate whether stabilizing sleep and/or events can improve well-being in undergraduates.

Clinical Trial Registration: NCT02846077.

Key words: mental health; public health; sleep and stress; stress; intra-individual variability; social jet lag; sleep regularity; mood; well-being

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Introduction

Adverse effects of short sleep duration on health are well-documented in the scientific literature [1]. Emerging findings suggest that sleep timing (i.e. sleep onset and offset) and sleep regularity (consistency in sleep timing from one day to the next) may be just as important for health and performance outcomes as sleep duration. Although the body of research on consequences of mistimed sleep is large and growing [2–4], a recent systematic review on sleep regularity—labeled *intraindividual variability*—concluded that “this body of literature is still at its infancy” [5]. Consistent associations have been reported between irregular sleep and adverse mental and physical health outcomes, including higher BMI, weight gain, affective disorders, insomnia, and generally poor sleep despite inconsistent methodologies used in available studies [5, 6].

In college students, sleep patterns are often highly variable as many students for the first time establish their own schedules and develop independent social rhythms while also facing academic demands and personal challenges. Irregular sleep patterns in this population have been associated with reduced physical health, including increased blood pressure [7], poorer psychomotor performance [8], lower academic performance [9, 10], higher BMI [11], and weight gain [12]. Numerous studies have linked sleep regularity to mental health in college students, using self-reports/questionnaires or average metrics (i.e. within-individual standard deviation) to quantify sleep regularity. An early study by Taub and Hawkins found that self-reported irregular sleepers scored lower on several personality dimensions (e.g. sociability, self-acceptance) than regular sleepers, despite no differences in average sleep duration [13]. Similarly, students at risk for bipolar disorder had less regular sleep patterns based on four weeks of diary data [14], which was confirmed by another study using seven days of objective actimetry recordings [15]. Self-reported irregular sleepers have fewer positive mood states and greater negative affect [8], as well as lower self-reported mental health scores and higher depression ratings [7, 16]. A recent study examining depressed mood and suicidal ideation at three time points over the course of 21 days in undergraduates at high-suicide risk found that irregular sleep patterns at baseline predicted suicidal and depressive symptoms at 7-day and 21-day follow-ups, and that sleep irregularity was a stronger predictor of acute suicidal ideation than depression severity [17]. Previous research has studied the effects of class schedules on college students’ alcohol consumption and academic performance [18, 19], assigning sleep timing a mediating role [20, 21]. Hershner and Chervin named “variable class schedules from day to day” a challenge in college students for good sleep hygiene and consequently for learning and mood [22]. To the best of our knowledge, no prior studies have examined the relationship between the regularity of sleep vs. event schedules and their relationship with well-being in undergraduates.

All of these studies used self-reports/questionnaires or average metrics (i.e. within-individual standard deviation) to quantify sleep regularity. In fact, the overwhelming majority of studies on sleep regularity, including all 53 studies in the systematic review by Bei and colleagues [5], used metrics that quantify overall variability in sleep after averaging across days, rather than quantifying the degree to which sleep patterns differ between consecutive days (i.e. on a circadian time-scale). However, rapid day-to-day changes in sleep patterns are

important to quantify because they would be expected to cause misalignment between the circadian system and sleep–wake cycles, since the circadian clock takes time to adjust to schedule changes. Such misalignment may be an underlying mechanism for the association between irregular sleep patterns and adverse health and performance outcomes. Metrics that quantify overall regularity, such as within-individual standard deviations or the widely used interdaily stability [23], do not specifically capture day-to-day (circadian) changes: e.g. if sleep times in a time series were randomly re-ordered, these metrics would return the identical value, even though the sleep regularity for a given interval within that time series, e.g. sleep on days 3–5, may have dramatically changed.

Two metrics have been recently developed to specifically capture day-to-day changes in sleep patterns: the Composite Phase Deviation (CPD [24]) and the Sleep Regularity Index (SRI [9]). The CPD metric combines sleep irregularity and sleep mistiming by quantifying (i) how different the midsleep times are compared to those on the previous day, and (ii) how far away midsleep times occur from an individual’s preferred sleep timing (*chronotype*, as measured by midsleep time on weekends [25]). The SRI calculates the probability of an individual being in the same state (asleep vs. awake) at any two time-points 24 h apart, scaled to values between 0 (completely irregular) and 100 (completely regular). Although CPD and SRI both capture day-to-day changes in sleep patterns, CPD may be complementary to SRI due to its composite nature, combining features of irregularity with mistiming. For example in shift work, sleep during the daytime for several consecutive night shifts tends to be highly regular (as captured by SRI and the irregularity component of the CPD metric) but as it occurs during the day it is largely mistimed for the majority of shift workers [26]. This mistiming is captured by the second component of the CPD metric but not by the SRI.

Using the recently developed CPD metric to specifically capture day-to-day changes in sleep patterns and event schedules, we examined the relationships among mistiming/irregularity of sleep schedules, event schedules, and self-reported well-being in 223 US college students studied during approximately 30 days. We formulated three study aims:

- (1) We aimed to test whether CPD for sleep is associated with well-being in college students and hypothesized that high-CPD (mistimed/irregular) sleep patterns would be associated with poor well-being upon awakening, on both a daily and an average (entire month) basis.
- (2) Previous studies have shown that sleep regularity can be experimentally manipulated by enforcing regular sleep schedules, yet found inconsistent results as to whether the manipulation leads to improvements in health- and performance-related outcomes [27–29]. We aimed to quantify the relationship between the timing and regularity of students’ sleep schedules and the timing and regularity of their event schedules, using continuous regression analysis and cluster analysis. We hypothesized that CPD for sleep and event schedules would be positively associated, resulting in two main clusters, i.e. mistimed/irregular sleepers on mistimed/irregular event schedules and regular/aligned sleepers on regular/aligned event schedules.
- (3) We aimed to quantify whether sleep and event schedules would have combined effects on well-being and

hypothesized that event schedules would exacerbate the effects of sleep, i.e. mistimed/irregular sleepers on mistimed/irregular event schedules would report the poorest well-being.

Although we did not formulate explicit hypotheses regarding the daily and average effects of sleep duration and sleep timing, both variables were included in the analysis to test that any potential impact of CPD was independent from these aspects of sleep. Hypotheses were not formulated for each well-being scale.

Methods

Participants and study protocol

Daily well-being and actimetry data were collected for approximately 30 days from 223 fulltime undergraduates at one midsize private Massachusetts university during fall and spring semesters between 2013 and 2016. The approximately 30 days of data collection began within the first weeks of start of semester so that they would end before a scheduled multiday vacation (e.g. thanksgiving holiday or spring break). Participants were excluded if pregnant or traveling more than one time zone 1 week before or during the study. Students were aged 18–27 (mean $\pm$ sd, 19.4 ± 1.5 y) and 37% were females ($n = 83$).

Students wore the actigraphy device Motionlogger-L (Ambulatory Monitoring, Inc., Ardsley, NY) on the wrist of their nondominant hand for approximately 30 days (6–34 days, median 29 days). Students also completed a daily online diary after awakening that included self-reports of: (i) sleep onset and offset times that were later used to assist determination of sleep onsets and offsets from actigraphy; (ii) the time of their first scheduled event (FSE, note: the event could be any academic, exercise, or extracurricular activity including meeting a friend); and (iii) their well-being on five visual nonnumerical scales (described below under Well-being section). Figure 1a illustrates the study design, showing daily sleep episodes, FSEs, and morning assessments of well-being for 30 days for one college student.

The study was in adherence with the Declaration of Helsinki and approved by the Committee on the Use of Humans as Experimental Subjects (Couhes) at Massachusetts Institute of Technology. The study was registered on ClinicalTrials.gov: NCT02846077. All participants provided written informed consent.

Data processing and study variables

Well-being

Students reported their well-being online on five visual analog (nonnumerical) scales every morning. Seventy-five percent of entries were within 3 h after awakening; entries more than 6 h after awakening (6%) were excluded from this analysis. The five scales *Sleepy–Alert*, *Sad–Happy*, *Sluggish–Energetic*, *Sick–Healthy*, and *Stressed–Calm* were later scored 0–100 with higher scores representing better well-being.

Actigraphy

Sleep onsets and offsets were determined using a combination of actigraphy and online sleep diaries [30, 31]. Students reported any actiwatch removals, and these were marked as missing data.

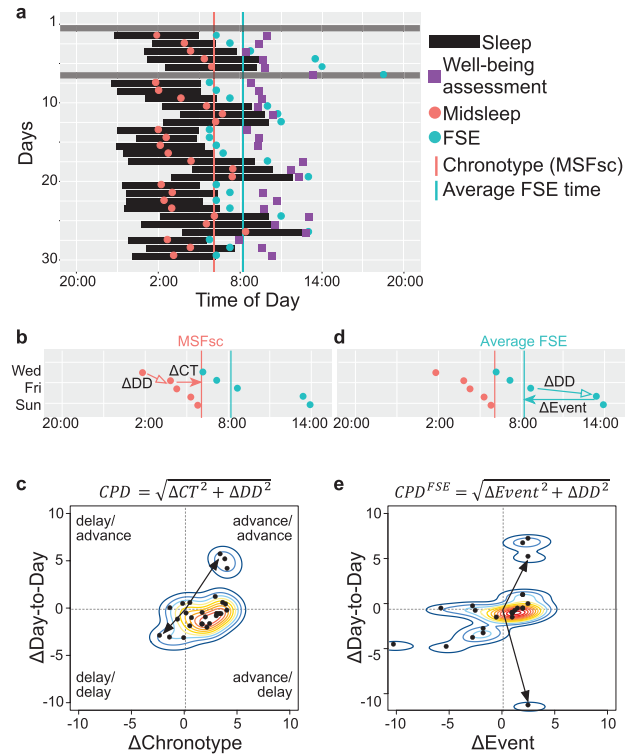


Figure 1. Example Composite Phase Deviation for sleep (CPD^{Sleep}) and for first scheduled event (FSE) timing (CPD^{FSE}). (a) Raster plot of one individual showing daily sleep episodes, FSEs, and well-being assessments over 30 days. Days 1 and 7 are missing sleep data (gray bars from left to right). MSFsc = chronotype (midsleep on weekends, corrected for sleep loss on weekdays). Panels b and d show an enlarged section of panel a with only midsleep and FSE information. (b) Enlarged section of panel a (days 1–7) for midsleep times with Δ Chronotype (Δ CT) and Δ Day-to-Day (Δ DD). Note that days 1 and 7 are missing, resulting in missing CPD data. (c) CPD plot for sleep. The arrows exemplify vectors from the origin to a data point. The CPD value of this data point is quantified by the length of the corresponding vector. Colored contour lines connect areas of equal data point density. (d) As in panel b but for FSEs. (e) As in panel c but for FSEs with Δ Event and Δ DD.

Chronotype

An individual's chronotype reflects how the circadian system embeds itself into the 24 h day with rhythms in physiology, cognition, and (sleep–wake) behavior occurring accordingly earlier or later [32]. Chronotype can be assessed from sleep–wake behavior using midsleep times on non-workdays to minimize the influence of external demands during the work week, such as forced wake-ups. Here, chronotype was calculated as the midpoint of the major sleep episode on weekends, corrected for over-sleeping due to sleep loss on weekdays (MSFsc) (equations 1–3) [25, 33]:

$$MSF = SON_{weekends} + \frac{1}{2} SDur_{weekends} \quad (1)$$

$$MSFsc = MSF - \frac{SDur_{weekends} - SDur_{weekdays}}{2}, \quad \text{if } SDur_{weekends} > SDur_{weekdays}, \quad (2)$$

$$\text{else } MSFsc = MSF \quad (3)$$

where MSF is midsleep on weekends; $SON_{weekends}$ sleep onset on weekends; $SDur_{weekends}$ sleep duration on weekends; $SDur_{weekdays}$ sleep duration on weekdays; MSFsc midsleep on weekends,

corrected for sleep loss on weekdays (chronotype); $SDur_{\text{weekly}}$ denotes weighted weekly average of sleep duration. We chose to use midsleep on weekends rather than nonevent days, because (i) 25% of students had 4 or less nonevent days per 30 days and we wanted to maximize the number of days used to calculate chronotype; (ii) irrespective of scheduled events, weekdays are usually socially different from weekends, and we aimed to increase comparability. This means that for some students the chronotype calculation included days with scheduled events, but we found fewer events were scheduled on weekends (37%) vs. weekdays (93%), and they began systematically later on weekends: median FSE time on weekends was 12:00 with 50% of events between 10:00 and 15:00, whereas median FSE time on weekdays was 10:00 with 50% of events between 9:30 and 11:00.

Composite Phase Deviation using midsleep times (CPD^{Sleep})

The CPD^{Sleep} metric quantifies day-to-day changes in sleep (irregularity component) and the extent of sleeping at the wrong time (mistiming component). The latter assumes that sleep during an internal sleep window provided by the circadian clock is optimal and sleeping outside or misaligned with this window is considered mistimed. The internal sleep window can be estimated by an individual's chronotype and MSFsc is therefore used as the reference to quantify mistimed sleep [24]. CPD^{Sleep} calculates how far away (in hours) sleep occurs from (i) the individual's chronotype ($\Delta\text{Chronotype}$), and (ii) the previous sleep episode ($\Delta\text{Day-to-Day}$). The $\Delta\text{Chronotype}$ (ΔCT) component reflects the mistiming of sleep (i.e. whether sleep occurs close to its optimal time), whereas $\Delta\text{Day-to-Day}$ (ΔDD) reflects irregularity of sleep timing (Figure 1b) (equations 4 and 5). CPD^{Sleep} plots have ΔCT on the horizontal axis and ΔDD on the vertical axis. In these plots (Figure 1c), the origin represents an "ideal state," where sleep occurs at the individual's optimal sleep time (chronotype) and at the same time every day. The deviation of any data point from the origin is quantified by the length of its corresponding vector; the CPD^{Sleep} metric is the average vector length (Figure 1c) (equation 6). For example in Figure 1c, a few data points are gathered in the upper-right quadrant of the plot, where sleep is advanced relative to chronotype (positive ΔCT) and advanced relative to the previous sleep episode (positive ΔDD). These data points are from Sundays to Mondays: after a delay in sleep times over the weekend, sleep is advanced due to class or other scheduled events. Missing midsleeps, such as from an "all-nighter," result in missing CPD^{Sleep} data (10% of days had no sleep data).

$$\Delta\text{Chronotype}_t (\Delta\text{CT}_t) = \text{MSFsc} - \text{Midsleep}_t \quad (4)$$

$$\Delta\text{Day-to-Day}_t (\Delta\text{DD}_t) = \text{Midsleep}_{t-1} - \text{Midsleep}_t \quad (5)$$

$$CPD_t^{\text{Sleep}} = \sqrt{\Delta\text{CT}_t^2 + \Delta\text{DD}_t^2} \quad (6)$$

where subscript t denotes a given day in the time series.

Composite Phase Deviation using FSE times (CPD^{FSE})

We applied the CPD approach described above to FSE times to assess the mistiming and irregularity of students' event schedules. CPD^{FSE} quantifies how far an event occurs (i) from the average event start time (ΔEvent), and (ii) from the previous event

($\Delta\text{Day-to-Day}$) (Figure 1d). Accordingly, missing events result in missing CPD^{FSE} data (21% of days had no FSE). ΔEvent and $\Delta\text{Day-to-Day}$ are then plotted against each other (Figure 1e). CPD^{FSE} is quantified using the length of the vector from the origin (i.e. both perfectly aligned and regular events) to each data point.

Statistics

All variables were tested for normality of distribution. Sleep duration was the only normally distributed variable (Shapiro-Wilk $W = 0.99, p = 0.45$). We therefore used nonparametric tests. Rank correlations (Spearman's rho) were computed to test associations between well-being and sleep variables. Mann-Whitney U tests were used for two-group comparisons (males vs. females) of well-being. Effect size r was calculated as $Z/\sqrt{N}$ for sex differences. Kruskal-Wallis tests were computed for more than two-group comparisons among chronotype categories (moderate: $\text{MSFsc} < 5:00$, late: $5:00-7:00$, very late: $>7:00$) and among clusters (outcome variables: average sleep duration, MSFsc , standard deviations of midsleep and scheduled event times, average time of FSE, average well-being; note: cluster analysis is described below). Effect size ϵ^2 was computed for cluster differences: $0.01-0.08$ small effect, $0.08-0.26$ medium effect, ≥ 0.26 large effect. A χ^2 -test was performed to examine sex distributions by cluster. Circular means and standard deviations were calculated for midsleeps and FSEs. In order to use linear statistics, midsleep was linearized by transforming midsleeps between 20:00 and 24:00 into values between -4.00 and 0.00 . Significance level was set to $\alpha = 0.05$. No meaningful auto- or cross-correlation components were detected in the time series and data were found to be stationary.

To check the appropriateness of linear regression vs. logistic regression, Q-Q plots and Shapiro-Wilk tests were used to test for normal distribution of model residuals; no violations were detected. Residuals vs. fitted values-plots and Breusch-Pagan tests were computed to test for heteroscedasticity; results were nonsignificant. Because 11% of all observations were missing data ($n = 759$, missing sleep or well-being or both on a given day) with a range of 0%–81% among students (median = 6.67%, $\text{IQR} = 3.33\%-10\%$), we conducted sensitivity analyses, excluding students with more than 10% of missing data (remaining sample: $n = 173$ students and 5289 observations).

Regression analysis

We were interested in both daily (within-person) and average (between-person) effects of CPD^{Sleep} on students' well-being. To disaggregate the two types of effects (between/average vs. within/daily), we computed a longitudinal growth model with time-varying covariates (TVCs) [34]. After checking that the TVCs in our dataset (CPD^{Sleep} , sleep duration, midsleep) were unrelated to time, we followed the traditional approach [35] to disaggregate between-person and within-person effects by calculating the person-specific mean ($\bar{z}_i$) and daily deviations from the person-specific mean ($\hat{z}_{ti}$) for all three TVCs. Both $\bar{z}_i$ and $\hat{z}_{ti}$ were then used as predictors in a random-intercept model:

$$Y_{ti} = (Y_{00} + Y_{01}\bar{z}_i + Y_{10}\hat{z}_{ti}) + (r_{ti} + u_{0i}) \quad (7)$$

where y_{00} is intercept (or grand mean); y_{01} direct estimate of the between-person effect (average effects); y_{10} direct estimate

of the within-person effect (daily effects); r_{ti} residual term (i.e. time-specific deviation from person-specific mean); u_{0i} is random residual term (i.e. unexplained (“random”) differences among individuals).

The model thus captures the relation between average levels of, e.g. CPD^{Sleep} and average levels of well-being across all individuals (via the estimate y_{01}). It also captures the relation between a given student’s daily deviation in CPD^{Sleep} (relative to the overall level of CPD^{Sleep}) and the student’s daily well-being (via the estimate y_{10}). We report the unstandardized coefficient b and standard errors in the main text, and provide full model information in [Supplementary Table S1A–E](#). To test for combined effects of sleep and event schedules on well-being, the interaction term $CPD^{Sleep} \cdot CPD^{FSE}$ was included both for daily and average effects. Two additional models were calculated separately for the two CPD^{Sleep} components to determine whether potential effects of CPD^{Sleep} were driven by sleep mistiming (ACT) vs. sleep irregularity (ADD), using absolute values of ΔCT and ΔDD .

The same type of longitudinal growth model with TVCs was computed to test daily and average effects of event schedules (CPD^{FSE}) (i) on sleep variables (CPD^{Sleep} , sleep duration, and midsleep), and (ii) on well-being scores of all five scales. All models were adjusted for sex, whereas age was not included due to its narrow range (75% of students were between 18 and 20). Study year was not included as a covariate, since it did not improve model AIC. To specifically examine the relationship between CPD^{FSE} and CPD^{Sleep} , we furthermore conducted cluster analysis (see below for details).

Regular linear models were used to compare average well-being among clusters with cluster, average sleep duration, chronotype (MSFsc), and sex in the same model. Methods for clustering are described below. For these models, reference group was Cluster 4: mistimed/irregular sleepers (high- CPD^{Sleep}) on mistimed/irregular event schedules (high- CPD^{FSE}). The multiplicative term $cluster \cdot chronotype$ was added to test for interactions.

Cluster analysis

We performed divisive hierarchical clustering in R (DIVISive ANALysis, DIANA [36]) to examine the relationship between sleep schedules (CPD^{Sleep}) and event schedules (CPD^{FSE}). Hierarchical clustering is an alternative approach to partitioning clustering (e.g. k -means clustering); it does not require the optimal number of clusters to be specified a priori. Agglomerative hierarchical clustering is a bottom-up approach starting with as many clusters as observations whereas DIANA is a top-down approach. In DIANA, all observations are initially in one cluster, the observation with maximum average dissimilarity is then moved to a new cluster, and this process is iterated until every observation is in a separate cluster. Divisive clustering is good at detecting large clusters, which is why we chose to perform DIANA. Results were compared with partitioning clustering using k -medioids. CPD^{Sleep} was positively skewed and thus log-transformed. Log-transforming CPD^{FSE} resulted in a heavily skewed distribution; CPD^{FSE} was thus not log-transformed. Both CPD^{Sleep} and CPD^{FSE} were scaled (mean = 0, sd = 1). R packages “cluster” [37], “factoextra” [38], and “fpc” [39] were used to perform the cluster analyses. The Jaccard Index was calculated to assess stability of clusters using the clusterboot function in

R. Clusters with a stability value less than 0.6 should be considered unstable; values between 0.6 and 0.75 indicate that the cluster is measuring a pattern in the data with moderate certainty about which points should be clustered together; values above 0.85 can be considered highly stable.

Results

Later chronotype is associated with more mistimed/irregular sleep but not with poorer well-being

The average chronotype (MSFsc) was $6:35 \pm 1.20$ h (interquartile range = 5:49–7:27). Only six students had a chronotype of MSFsc < 4:00. Cut-offs for chronotype categories as shown in [Table 1](#) were therefore chosen as: moderate types (MSFsc < 5:00), late types (5:00–7:00), and very late types (>7:00), in line with previous studies [25, 40].

Later chronotype (MSFsc) was associated with higher CPD^{Sleep} ($r = 0.48$, $p < 0.001$) and younger age ($r = -0.15$, $p = 0.03$) ([Figure 2a](#) and [b](#)). Males had marginally later chronotypes than females ($p = 0.06$). No significant correlations were observed between chronotype and sleep duration ($r = 0.07$, $p = 0.33$) or chronotype and any of the five scales ($-0.002 < r < -0.09$, $p > 0.17$). Age was not associated with any of the five scales ($-0.06 < r < 0.04$), CPD^{Sleep} ($r = -0.10$), or sleep duration ($r = 0.07$), all $p > 0.33$. The five well-being scales were correlated with each other, with coefficients ranging from $r = 0.56$ (Sick–Healthy and Stressed–Calm) to $r = 0.90$ (Sleepy–Alert and Sluggish–Energetic), all $p < 0.001$ ([Supplementary Table S2](#) and [Figure S1](#)).

Although sex was not associated with CPD^{Sleep} (Mann–Whitney U , $r = -0.12$, $p = 0.33$) or sleep duration ($r = 0.06$, $p = 0.69$), males did report significantly higher (i.e. “better”) scores on four of the five scales ([Table 1](#), [Figure 2c–f](#)): Sleepy–Alert ($r = -0.16$, $p = 0.02$), Sad–Happy ($r = -0.18$, $p = 0.006$), Sluggish–Energetic ($r = -0.16$, $p = 0.02$), and Stressed–Calm ($r = -0.30$, $p < 0.001$). Males and females did not significantly differ on the scale Sick–Healthy ($r = -0.09$, $p = 0.15$).

Weak to moderate correlations were observed between percentage of missing data (11% overall, range 0–81%, 75%-quartile = 10%) and CPD^{Sleep} ($r = 0.19$, $p = 0.004$) or chronotype ($r = 0.19$, $p = 0.004$). No significant correlations with percentage of missing data were observed for sleep duration ($r = 0.05$, $p = 0.42$) or any of the five scales ($-0.01 < r < 0.05$, $p > 0.46$).

Composite Phase Deviation (CPD^{Sleep}) and sleep duration predict college students’ well-being on different timescales (Hypothesis 1)

Daily well-being (within-person effects)

Daily sleep duration was the strongest significant predictor of daily well-being on four of five scales: with every additional hour of sleep, daily Sleepy–Alert, Sad–Happy, Sluggish–Energetic, and Stressed–Calm improved by 2.6, 0.5, 1.6, and 1.0 units, respectively, on a scale from 0 (poor) to 100 (good) ([Table 2](#)). Predictive power of sleep duration was highest for daily Sleepy–Alert, whereas sleep duration had no significant effect on daily Sick–Healthy. Midsleep predicted Stressed–Calm on a daily basis: with every hour that midsleep was later, students reported feeling calmer by 0.7 units. CPD^{Sleep} was not a significant predictor of any of the five scales on a daily basis.

Table 1. Sample demographics and study variables in the total sample, by sex and chronotype categories

	Total sample (n = 223)	Females (n = 83)	Males (n = 140)	Moderate chronotype (<5:00, n = 19)	Late chronotype (5:00–7:00, n = 122)	Very late chronotype (>7:00, n = 82)
Age (years)	19.4 ± 1.5 (18–27)	19.4 ± 1.5 (18–27)	19.4 ± 1.5 (18–27)	19.6 ± 1.0 (18–22)	19.4 ± 1.5 (18–27)	19.4 ± 1.4 (18–24)
Sex (% female (n))	37% (83)	100% (83)	0% (0)	46% (6)	79% (54)	39% (23)
Sleep duration (h)	6.8 ± 0.78 (5.2–9.1)	6.8 ± 0.7 (5.2–8.6)	6.8 ± 0.8 (5.2–9.1)	6.8 ± 0.7 (5.3–8.1)	6.8 ± 0.7 (5.3–9.1)	6.8 ± 0.8 (5.2–8.6)
Chronotype (MSFsc) (h)	6:35 ± 1.2 (3:29–10:43)	6:26 ± 1.2 (3:52–10:43)	6:40 ± 1.2 (3:29–9:02)	4:20 ± 0.5 (3:29–4:57)	6:05 ± 0.5 (5:01–6:58)	7:50 ± 0.6 (7:01–10:43)
*CPD ^{Sleep} (h)	1.8 ± 0.6 (0.8–4.0)	1.7 ± 0.5 (0.8–3.5)	1.8 ± 0.6 (0.8–4.0)	1.5 ± 0.4 (0.9–2.5)	1.7 ± 0.6 (0.8–4.0)	2.1 ± 0.5 (1.0–3.6)
CPD ^{FSE} (h)	2.4 ± 1.3 (0.00–6.0)	2.6 ± 1.2 (0.00–6.0)	2.2 ± 1.4 (0.1–5.6)	2.5 ± 1.2 (0.7–4.3)	2.3 ± 1.3 (0.00–5.6)	2.5 ± 1.4 (0.00–6.0)
*Sleepy–Alert (0–100)	50.6 ± 18.6 (5.7–95.8)	47.1 ± 17.2 (16.4–89.4)	52.8 ± 19.2 (5.7–95.8)	55.9 ± 17.0 (26.6–84.8)	50.7 ± 18.6 (5.7–94.6)	49.3 ± 19.0 (10.4–95.8)
*Sad–Happy (0–100)	60.9 ± 15.8 (14.8–97.8)	57.7 ± 13.7 (22.2–94.8)	62.9 ± 16.6 (14.8–97.8)	64.3 ± 13.5 (44.9–81.9)	60.5 ± 16.0 (14.8–97.8)	60.8 ± 15.9 (22.2–96.2)
*Sluggish–Energetic (0–100)	51.1 ± 17.6 (5.3–95.0)	47.7 ± 15.8 (14.6–92.9)	53.1 ± 18.4 (5.3–95.0)	55.0 ± 15.7 (32.4–81.3)	51.6 ± 17.7 (5.3–93.7)	49.3 ± 18.0 (7.9–95.0)
Sick–Healthy (0–100)	64.6 ± 18.4 (4.3–100.0)	62.3 ± 17.5 (4.3–97.9)	66.0 ± 18.9 (22.3–100.0)	63.2 ± 20.1 (22.4–90.6)	65.2 ± 18.1 (22.3–100.0)	64.1 ± 18.7 (4.3–100.0)
*Stressed–Calm (0–100)	53.1 ± 18.4 (3.1–96.2)	46.5 ± 16.1 (14.1–92.7)	57.0 ± 18.5 (3.1–96.2)	55.8 ± 14.1 (32.5–81.2)	53.2 ± 18.4 (3.1–96.2)	52.4 ± 19.2 (10.8–95.7)

Chronotype categories were chosen based on previous studies [25, 40]. Mean values ± sd (range) are shown. MSFsc = midsleep on weekends, corrected for sleep loss on weekdays. CPD^{Sleep} = Composite Phase Deviation using midsleeps. CPD^{FSE} = Composite Phase Deviation using first scheduled event (FSE) times. Well-being scales range from 0 to 100, with higher values representing better well-being.

*Significantly different among chronotype groups (Kruskal–Wallis, $p < 0.001$).

*Significantly different between males and females (Mann–Whitney U, $p < 0.05$).

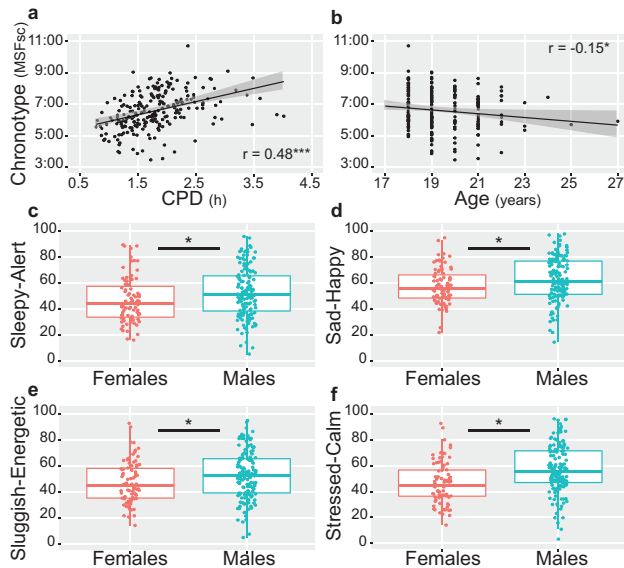


Figure 2. Associations of well-being and CPD^{Sleep} with chronotype and sex. A late chronotype (MSFsc) was associated with (a) higher Composite Phase Deviation (CPD^{Sleep}) and (b) younger age. Males scored higher (“better”) on (c) Sleepy–Alert, (d) Sad–Happy, (e) Sluggish–Energetic, and (f) Stressed–Calm. r = rank correlation coefficient Spearman’s rho. Sex comparisons in panels c–f are based on nonparametric Mann–Whitney U tests. Horizontal lines denote significant group differences: * $p < 0.05$, *** $p < 0.001$.

Average well-being (between-person effects)

In contrast, average CPD^{Sleep} was the strongest significant predictor of average (across ~30 days) well-being on three of five

scales: with every additional hour of CPD^{Sleep} (i.e. more mistimed/irregular sleep), Sad–Happy, Sick–Healthy, and Stressed–Calm worsened by 5.6, 5.9, and 6.2 units, respectively (Table 2). Predictive power of CPD^{Sleep} was highest for average Stressed–Calm, whereas CPD^{Sleep} had no effect on average ratings of Sleepy–Alert and Sluggish–Energetic. Average sleep duration was a significant predictor of average Stressed–Calm, which improved by 3.8 units with every additional hour of sleep. Average midsleep had no effect on average well-being on any of the five scales.

Given that the CPD^{Sleep} metric combines two components—(i) the sleep mistiming component Δ Chronotype (Δ CT) and (ii) the sleep irregularity component Δ Day-to-Day (Δ DD)—we were interested in their individual predictive contributions to well-being. We therefore re-computed the longitudinal growth model replacing CPD^{Sleep} with Δ CT or Δ DD. Δ DD (irregularity), but not Δ CT (mistiming), was a significant predictor for average well-being on four of five scales: Sad–Happy, Sluggish–Energetic, Sick–Healthy, and Stressed–Calm. With every additional hour of day-to-day irregularity (Δ DD), students reported feeling overall less happy ($b = -9.1$, $SE = 2.8$, $p = 0.002$), less energetic ($b = -7.0$, $SE = 3.3$, $p = 0.03$), less healthy ($b = -8.1$, $SE = 3.5$, $p = 0.02$), and less calm ($b = -9.6$, $SE = 3.2$, $p = 0.002$). Δ CT (mistiming) was a significant predictor on the scale Stressed–Calm: with every additional hour of mistimed sleep, students reported feeling less calm ($b = -6.2$, $SE = 3.0$, $p = 0.04$). Daily Δ CT or Δ DD were not significant predictors of daily well-being (all $p > 0.60$).

Results of the longitudinal growth models did not change when variables were entered stepwise (beginning with CPD^{Sleep}). To test the influence of missing data, we excluded students with more than 10% of missing data and conducted a sensitivity analysis with the remaining sample ($n = 173$ students with 5289

Table 2. Longitudinal growth model with time-varying covariates (random intercept)

	Sleepy-Alert		Sad-Happy		Sluggish-Energetic		Sick-Healthy		Stressed-Calm	
	Estimate	SE	Estimate	SE	Estimate	SE	Estimate	SE	Estimate	SE
Intercept	70.1	13.9	54.9	11.3	69.5	13.0	67.2	13.6	31.5	12.8
CPD ^{Sleep} (h, daily)	-0.03	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.1	0.2
Sleep duration (h, daily)	2.6***	0.2	0.5***	0.2	1.6***	0.2	0.1	0.2	1.0***	0.2
Midsleep (h, daily)	0.4	0.2	0.2	0.2	-0.1	0.2	-0.3	0.2	0.7***	0.2
CPD ^{Sleep} (h, average)	-2.6	2.5	-5.6**	2.0	-3.2	2.3	-5.9*	2.4	-6.3**	2.3
Sleep duration (h, average)	-1.8	1.8	1.9	1.5	-1.2	1.7	0.5	1.8	3.8*	1.6
Midsleep (h, average)	-0.9	1.2	-0.02	0.9	-1.2	2.5	0.2	1.2	-0.01	1.1

Person-specific means and daily deviations from person-specific mean were calculated to estimate between-person (average) and within-person (daily) effects of sleep variables on well-being. An unstandardized regression coefficient of e.g. $b = -5.6$ for CPD^{Sleep} means that with every additional hour of mistimed/irregular sleep, well-being worsens by 5.6 units on a scale from 0 (poor) to 100 (good). CPD^{Sleep} = Composite Phase Deviation using midsleeps. Estimate = unstandardized regression coefficient b . SE = standard error. All estimates are sex-adjusted.

** $p < 0.01$.

*** $p < 0.001$.

observations, 22% excluded data). Results were the same for both daily and average well-being scores.

Event schedules (CPD^{FSE}) are associated with sleep schedules (CPD^{Sleep}) on a daily but not average basis (Hypothesis 2)

Daily basis (within-person effects)

CPD^{FSE} was associated with CPD^{Sleep}, sleep duration, and midsleep on a daily basis: with every additional hour of CPD^{FSE} (i.e. more mistimed/irregular events), CPD^{Sleep} increased by 2.4 min ($b = 0.04$, $SE = 0.01$, $p < 0.001$), sleep duration decreased by 2.4 min ($b = -0.04$, $SE = 0.01$, $p < 0.001$), and midsleep delayed by 3.0 min ($b = 0.05$, $SE = 0.01$, $p < 0.001$), indicating that event schedules are more variable than sleep schedules (e.g. a difference of 5 h in CPD^{FSE} results in a difference of 12 min in CPD^{Sleep} and sleep duration and 15 min in midsleep).

Average basis (between-person effects)

CPD^{FSE} was not associated with CPD^{Sleep}, sleep duration, or midsleep on an average level (all $b < 0.08$, all $p > 0.19$). We furthermore specifically checked that there was no relationship between CPD^{FSE} and chronotype: event schedules of late types were on average not more mistimed/irregular than event schedules of earlier types ($b = 0.02$, $SE = 0.06$, $p = 0.75$).

Using hierarchical cluster analysis, the DIANA dendrogram (a tree diagram showing the number of clusters at different levels) identified approximately equally partitioned two-cluster and four-cluster solutions, when cut close to the top (Figure 3a). If there were a relationship between event schedules and sleep schedules, e.g. such that high-CPD event schedules promote high-CPD sleep schedules, we would expect higher density of data points in the upper-right (sleep and event schedules both high-CPD) and bottom-left (sleep and event schedules both low-CPD) quadrants, along the diagonal axis of the cluster plot. The two-clusters solution (Figure 3b) instead separated the data along the vertical axis into low-CPD^{FSE} (aligned/regular event schedules) and high-CPD^{FSE} (mistimed/irregular event schedules) clusters, illustrating that mistimed and irregular sleep occurs on either type of event schedule. This is consistent with the lack of association between CPD^{FSE} and CPD^{Sleep} on an average level.

The four-clusters solution (Figure 3c) further split the low- and high-CPD^{FSE} clusters along the horizontal axis into low- and

high-CPD^{Sleep} clusters. The four resulting clusters can thus be characterized as: aligned/regular sleepers on aligned/regular schedules (low-CPD^{Sleep}/low-CPD^{FSE}) (Cluster 1, $n = 48$), aligned/regular sleepers on mistimed/irregular schedules (low-CPD^{Sleep}/high-CPD^{FSE}) (Cluster 2, $n = 73$), mistimed/irregular sleepers on aligned/regular schedules (high-CPD^{Sleep}/low-CPD^{FSE}) (Cluster 3, $n = 61$), and mistimed/irregular sleepers on mistimed/irregular schedules (high-CPD^{Sleep}/high-CPD^{FSE}) (Cluster 4, $n = 41$). Other differences include earlier chronotypes in Clusters 1 and 2 (aligned/regular sleepers) than in Clusters 3 and 4 (mistimed/irregular sleepers) (Kruskal-Wallis, $p < 0.001$, $\epsilon^2 = 0.20$) (Figure 3d). Age and sex distributions did not differ among clusters (Kruskal-Wallis, $p = 0.31$, respectively χ^2 , $p = 0.17$). Figure 3e-h shows exemplary sleep and event schedules for one individual from each cluster (note that these individuals were selected to show individual differences as clearly as possible). Comparison with partitioning clustering using k -medioids as well as excluding students with more than 10% missing data yielded virtually identical clusters. The Jaccard Index yielded values of 0.71 (Cluster 1), 0.77 (Cluster 2), 0.70 (Cluster 3), and 0.76 (Cluster 4), indicating that all clusters were moderately stable, with Clusters 2 and 4 being slightly more stable than Clusters 1 and 3.

Poor well-being reported by mistimed/irregular sleepers (high-CPD^{Sleep}) is exacerbated by mistimed/irregular event schedules (high-CPD^{FSE}) (Hypothesis 3)

To test for combined effects of sleep and event schedules on well-being, we included the interaction term CPD^{Sleep}*CPD^{FSE} for both daily and average effects. Whereas the interaction terms did not reach significance, the coefficients indicated that the combination of high-CPD^{Sleep} and high-CPD^{FSE} further lowered students' daily and average well-being on all five scales (daily/average: Sleepy-Alert: $b_{\text{interaction}} = -0.05/-0.48$; Sad-Happy: $b_{\text{interaction}} = -0.04/-1.67$; Sluggish-Energetic: $b_{\text{interaction}} = -0.07/-0.96$; Sick-Healthy: $b_{\text{interaction}} = -0.06/-1.66$; Stressed-Calm: $b_{\text{interaction}} = -0.06/-1.60$; all $p > 0.22$).

We also tested for combined effects of CPD^{Sleep} and CPD^{FSE} by comparing students' average well-being among clusters using linear regression models. Across scales, the poorest average well-being was reported by students who were on both mistimed/irregular sleep and event schedules (Cluster 4)

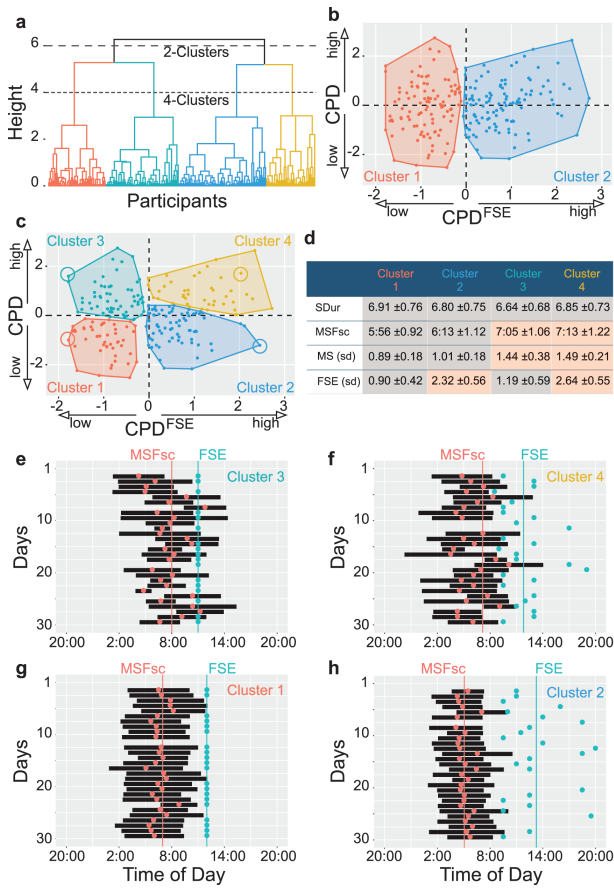


Figure 3. Cluster analysis. Divisive hierarchical clustering [36] was used to examine the relationship between sleep schedules and event schedules. Sleep schedules were assessed by Composite Phase Deviation using midsleeps (CPD^{Sleep}), whereas event schedules were assessed by Composite Phase Deviation using FSE times (CPD^{FSE}). (a) The dendrogram shows a two-clusters and a four-clusters solution, depending on where the dendrogram is cut. (b) The two-clusters solution groups the data into low- CPD^{FSE} (aligned and regular event schedules, Cluster 1) and high- CPD^{FSE} (mismatched and irregular event schedules, Cluster 2) clusters. Axes show z-scaled CPD^{Sleep} and CPD^{FSE} values, i.e. a value of -1 equals 1 sd below the sample mean. (c) The four-clusters solution further splits the data along the horizontal axis: aligned/regular sleepers on aligned/regular schedules (low- CPD^{Sleep} /low- CPD^{FSE}) (Cluster 1, $n = 48$), aligned/regular sleepers on mismatched/irregular schedules (low- CPD^{Sleep} /high- CPD^{FSE}) (Cluster 2, $n = 73$), mismatched/irregular sleepers on aligned/regular schedules (high- CPD^{Sleep} /low- CPD^{FSE}) (Cluster 3, $n = 61$), and mismatched/irregular sleepers on mismatched/irregular schedules (high- CPD^{Sleep} /high- CPD^{FSE}) (Cluster 4, $n = 41$). The four colored circles show the four individuals shown in panels e–h. (d) Characteristics of the four clusters by sleep duration (SDur), chronotype (MSFsc, sleep loss-corrected midsleep on weekends), standard deviation of midsleeps (MS (sd)), and standard deviation of first scheduled events (FSE (sd)). Colored boxes (gray and red) mark statistical differences between clusters (Kruskal–Wallis, $p < 0.05$). Effect sizes (ϵ^2) for cluster comparisons were as follows: $SD\epsilon^2 = 0.02$, $MSFsc \epsilon^2 = 0.20$, $MS (sd) \epsilon^2 = 0.60$, and $FSE (sd) \epsilon^2 = 0.67$. Raster plots are shown of one individual from each cluster (note that individuals were selected to illustrate differences): (e) Cluster 3, (f) Cluster 4, (g) Cluster 1, (h) Cluster 2. Black bars = sleep episodes. Red dots = midsleeps. Red line = chronotype (MSFsc, sleep loss-corrected midsleep on weekends). Blue dots = first scheduled events (FSEs). Blue line = average start time of FSE.

(Figure 4a–e). These students reported feeling significantly less happy ($b = -8.92$, $SE = 3.15$, $p = 0.005$), less energetic ($b = -7.53$, $SE = 3.55$, $p = 0.03$), less healthy ($b = -8.69$, $SE = 3.74$, $p = 0.02$), and less calm ($b = -10.82$, $SE = 3.53$, $p = 0.002$) than students who were also on mismatched/irregular schedules but whose sleep was (relatively) aligned and regular (Cluster 2). They also felt less calm/

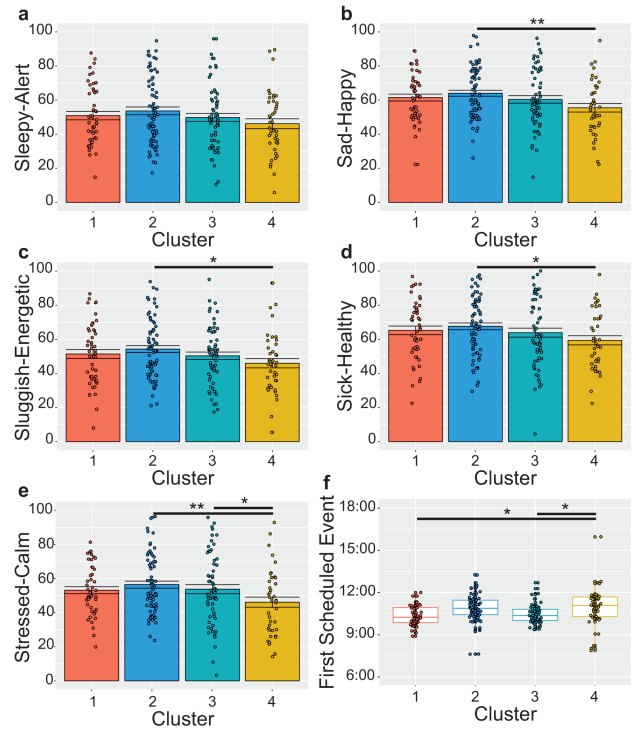


Figure 4. Well-being by clusters. Scores (mean ± SE) were compared among the four clusters on scales (a) *Sleepy–Alert*, (b) *Sad–Happy*, (c) *Sluggish–Energetic*, (d) *Sick–Healthy*, and (e) *Stressed–Calm*. Cluster 1: aligned/regular sleepers on aligned/regular event schedules (low- CPD^{Sleep} /low- CPD^{FSE}). Cluster 2: aligned/regular sleepers on mismatched/irregular event schedules (low- CPD^{Sleep} /high- CPD^{FSE}). Cluster 3: mismatched/irregular sleepers on aligned/regular event schedules (high- CPD^{Sleep} /low- CPD^{FSE}). Cluster 4: mismatched/irregular sleepers on mismatched/irregular event schedules (high- CPD^{Sleep} /high- CPD^{FSE}). Mismatched/irregular sleepers on mismatched/irregular event schedules (Cluster 4) reported the poorest well-being, whereas aligned/regular sleepers on mismatched/irregular event schedules (Cluster 2) reported the best well-being. The latter may be explained by later average start times of first scheduled events on mismatched/irregular event schedules (Clusters 2 and 4) compared to aligned/regular event schedules (Clusters 1 and 3), as shown in panel f. Horizontal lines denote significant group differences with * $p < 0.05$ and ** $p < 0.01$, derived from linear regression models with Cluster 4 as reference.

more stressed than students in Cluster 3 (Figure 4e), whose sleep was mismatched/irregular but who were on aligned/regular event schedules (*Stressed–Calm*: $b = -7.25$, $SE = 3.50$, $p = 0.04$), suggesting that for mismatched/irregular sleepers feeling stressed may decrease on aligned/regular event schedules. Effect sizes (ϵ^2) for comparisons of well-being among clusters were as follows: *Sleepy–Alert* $\epsilon^2 = 0.11$, *Sad–Happy* $\epsilon^2 = 0.13$, *Sluggish–Energetic* $\epsilon^2 = 0.12$, *Sick–Healthy* $\epsilon^2 = 0.13$, and *Stressed–Calm* $\epsilon^2 = 0.14$.

Contrary to our hypothesis, students with aligned/regular sleep and event schedules (Cluster 1) did not have the highest well-being scores; students in Cluster 2 whose sleep was aligned/regular but who were on mismatched/irregular event schedules reported (nonsignificantly) better well-being. We tested whether average FSE time differed among clusters as a potential explanation: FSEs started on average approximately 30–45 min later for students on irregular event schedules (Cluster 2: 10:55, Cluster 4: 11:00) than for students on regular event schedules (Cluster 1: 10:21, Cluster 3: 10:26) (Kruskal–Wallis, $p < 0.01$, $\epsilon^2 = 0.12$) (Figure 4f).

Well-being differences among clusters were not driven by chronotype (all $p > 0.28$): earlier chronotypes did not report

worse well-being than late chronotypes on mistimed/irregular event schedules, and among mistimed/irregular sleepers early chronotypes did not report worse well-being than late chronotypes.

Mistimed/irregular sleepers (Clusters 3 and 4) reported on average poorer well-being than aligned/regular sleepers (Clusters 1 and 2) (e.g. *Sad-Happy*: $b = -4.52$, $SE = 2.35$, $p = 0.02$), whereas well-being of students on mistimed/irregular event schedules (Clusters 2 and 4) did not differ from students on aligned/regular event schedules (Clusters 1 and 3), suggesting that sleep schedules may be more important for well-being than event schedules. This finding was confirmed using longitudinal growth models, yielding CPD^{Sleep} but not CPD^{FSE} as a significant predictor of average well-being: *Sad-Happy* (CPD^{Sleep} : $b = -5.78$, $p < 0.01$ vs. CPD^{FSE} : $b = -0.40$, $p = 0.63$), *Sick-Healthy* (CPD^{Sleep} : $b = -5.92$, $p = 0.02$ vs. CPD^{FSE} : $b = -0.59$, $p = 0.55$), and *Stressed-Calm* (CPD^{Sleep} : $b = -6.27$, $p < 0.01$ vs. CPD^{FSE} : $b = 0.40$, $p = 0.67$).

Discussion

In this study, we used a recently developed metric—the Composite Phase Deviation (CPD)—to quantify the mistiming and irregularity of sleep and FSEs in undergraduate students. This extension of CPD from its original application of sleep timing to other events generated novel insights into the relationship between sleep and event schedules and allowed us to quantify whether these schedules are predictive of self-reported well-being upon awakening, either at the daily level or the average level (across ~30 days). Our hypothesis that mistimed and irregular sleep patterns (i.e. high- CPD^{Sleep}) would be associated with poorer well-being was confirmed for average well-being but rejected for daily well-being. Contrary to our expectations, CPD for sleep and CPD for events were found to be (weakly) positively associated with one another at the daily level, and not at the average level in this population, i.e. generally mistimed and irregular event schedules were not associated with overall mistimed and irregular sleep. CPD^{Sleep} and CPD^{FSE} did interact on well-being, such that effects of mistimed/irregular sleep on well-being were exacerbated by mistimed/irregular event schedules.

A recent systematic review of sleep regularity [5] identified the need to study differential associations between various sleep dimensions and outcomes so as to develop a cohesive theoretical framework for the effect of sleep regularity. We report here such differential associations, namely for duration, timing, and regularity of sleep, as well as for different timescales (daily vs. average). Specifically, we conclude that CPD^{Sleep} is a useful predictor of inter-individual differences in average well-being, whereas sleep duration is a useful predictor of intra-individual daily variations in well-being relative to an individual's average. Moreover, we conclude that the relationship between CPD^{Sleep} and average well-being is driven by day-to-day differences in sleep timing (ADD) but not its misalignment (relative to an individual's preferred sleep timing, ΔCT). A study by Whiting and Murdock [41] found that consistently short sleep had adverse effects on attention whereas occasionally short sleep did not, supporting our finding that irregular sleep timing was associated with poorer well-being after some exposure time (~30 days). Although we cannot determine causality of these associations from these data, the findings suggest that chronic exposure to

irregular sleep patterns may be a specific risk factor for lowered average well-being.

Whereas some previous studies have associated later sleep timing (chronotype) with worse mood and well-being [42], we found no such association on an average level. This may be due to the absence of early chronotypes in this cohort; since the sleep timing in our sample of college students was overall delayed compared with the US population [25], we could only compare moderate and late chronotypes. On a daily level, midsleep timing was related to well-being on one scale: students reported feeling more calm (less stressed) when their midsleep was delayed. A later midsleep might indicate that no class or other events were forcing students to get up early, thus reducing feelings of stress.

Although the finding that average CPD^{Sleep} was associated with average well-being may indicate that irregular sleep-wake behavior could be trait-like (i.e. some individuals may tend to be irregular sleepers irrespective of circumstances), a recent study using a physiological mathematical model of sleep-wake regulation showed that irregular sleep-wake patterns can result from the interaction of endogenous characteristics (e.g. circadian period) with external factors (i.e. ability to control light-dark cycles) [43].

Our findings overall demonstrate that sleep and event schedules are very loosely coupled in these undergraduate college students. At the average level, there was no association between sleep and event schedules. At the daily level, the association was significant but weak; hours of difference in event times resulted in a difference of only minutes in sleep. This may be due to the late start times of most FSEs in this population, meaning they do not face the same sleep curtailment pressures as other populations, such as high school students with very early start times [44]. The type of FSE may also influence sleep (e.g. team athletic practice vs. class or extracurricular events). We do not know the type of FSE from these data; further studies can explore this. Since our analysis focuses on FSEs, we may also be overlooking impacts of other scheduled events on sleep, such as late social events that may delay sleep onsets (which were not collected in this sample). Similar to chronotype, our findings may also be influenced by the absence of individuals with very regular (rigid) schedules. Although some students were relatively regular compared to others, only $n = 11$ (5%) had $CPD^{FSE} < 1$ h. Extending this work to other cohorts that include more rigid work and/or sleep schedules would be predicted to reveal a stronger coupling between sleep timing and work constraints [43, 45].

Self-reported well-being has been shown to largely depend on mood [46], which in turn shows a circadian rhythmicity [47]. Here, we have examined the impact of preceding sleep on well-being upon awakening but it is also possible that well-being before bedtime might affect the following sleep episode. Future work could investigate (i) the impact of preceding sleep (CPD, duration, timing) on well-being depending on the time of the assessment (morning vs. evening) and (ii) the impact of evening well-being on the following sleep episode (CPD, duration, timing).

We did not see any effects of CPD^{Sleep} nor its components ΔCT (mistiming) and ΔDD (irregularity) on daily well-being. It is important to note that we used absolute values of ΔCT and ΔDD , thereby ignoring the direction of the deviation (i.e. advance vs. delay). Future analyses could examine daily effects of ΔCT and

ADD using relative values to distinguish between effects that are potentially different for advances vs. delays in sleep timing.

Among the limitations of this study is the fact that CPD^{Sleep} does not consider naps and treats nights with no sleep (all-nighters) as missing data, since CPD^{Sleep} is based on mid-sleep time of the major sleep episode. This may have resulted in an underestimation of the effect of irregular sleep/wake behavior on well-being. Other metrics, such as the SRI, are better equipped to deal with fragmented sleep/wake patterns. In addition, whereas the amount of missing data did not appear to impact the results based on sensitivity analyses (i.e. excluding those with the most missing data did not change the relationship between well-being and sleep), we cannot establish whether it impacted the calculation of CPD^{Sleep} and other sleep variables. It is plausible that extreme sleep (very short/early/rigid or very long/late/irregular) may be more likely to be missing data, e.g. students may take off their actiwatch or forget to fill out the sleep diary when there are special occasions (e.g. birthdays, parties, stressful events) that are more likely to result in atypical sleep. We therefore assume that the missing values in our dataset are missing not at random. There is no universal method to properly handle data that are missing not at random, and we therefore did not impute missing values. Although we may have biased toward the null and underestimated the relationships between mistiming/irregularity and other variables, it is important to note that missing data would affect both the estimates and the standard deviations of the estimates of those relationships [48]. Another limitation includes the fact that cluster analyses can result in unstable groups, and our finding of four clusters needs to be replicated in an external sample.

In their review, Bei et al. pointed out that no study justified the number of days to employ the sleep regularity method [5]. Future work should empirically determine in various populations the minimum number of days required for (i) reliable estimates of sleep regularity metrics, including CPD^{Sleep} and SRI; and (ii) reliable prediction of specific outcomes from these metrics. We found here that 30 days is sufficient to determine average well-being using CPD^{Sleep}, but it may be possible to use shorter time-spans. Sano et al. previously demonstrated, e.g. that SRI computed across 4–5 previous nights is sufficient to predict daily self-reported well-being [49]. Future work should perform a detailed comparison of CPD to other metrics of sleep regularity, including the SRI. Future studies could also use the CPD metric to assess variability in other dimensions of sleep, including sleep latency, sleep efficiency, and sleep fragmentation.

In conclusion, we find that mistimed/irregular sleep patterns are largely independent of mistimed/irregular FSEs in these college students; however, both of these factors are associated with worse average well-being over a period of approximately 30 days. Future work should extend this to other populations. These findings suggest that interventions to stabilize and align sleep and/or FSEs have potential to improve well-being.

Supplementary Material

Supplementary material is available at SLEEP online.

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